

Navneet Shankar

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EXPERIENCE

Fralin Life Sciences Institute – Virginia Tech

Blacksburg, VA

Software Engineer

Aug 2025 – Present

- Enabled research faculty and grants administrators to search and analyze **600K+** NIH/NSF award records across **58K+** institutions by building AWS-hosted microservices with a Next.js/TypeScript frontend, Go/Gin REST APIs, and Python/FastAPI ETL pipelines.
- Developed NIH and NSF ingestion pipelines covering **10+ years** of historical award data across **200 GB** of storage; implemented deterministic date-window splitting for NIH API pagination limits and SAVEPOINT-based PostgreSQL upserts for fault-isolated writes at **2,500+ records/day**.
- Implemented full-text search using PostgreSQL indexing, parallel Go routines, and Redis caching, delivering **sub-200 ms** search responses across awards and faculty profiles.
- Built a LangChain/LangGraph Statistics Agent that routes natural language queries through an agentic graph to return computed statistics and **15+ chart types** over the NIH/NSF award dataset.
- Shipped RAG-based semantic search with Pinecone and GraphRAG question answering with Neo4j, using bounded retrieval context to reduce hallucinations in generated responses.

Citi

Chennai, India

Software Engineer – Olympus (Big Data Division)

Sep 2022 – Jul 2024

- Engineered Apache Spark data pipelines in Scala and Python processing **3+ TB** daily, reducing ingestion time by **30%** through pipeline and storage optimizations.
- Reduced HDFS small-file sprawl in Hive-backed storage by implementing scheduled compaction workflows and partition-aware Spark write strategies, lowering per-partition file counts by **40%** and reducing NameNode overhead.
- Built and scaled Kafka-based stream processing services handling **3 TB** of data daily, powering real-time risk analytics dashboards with **sub-second** latency and monitoring aligned to a **99.9% SLA**.

Citi

Chennai, India

Software Engineer – Commercial Credit Cards

Jul 2021 – Sep 2022

- Built production REST microservices using Spring Boot in a **50+** service ecosystem with Netflix Eureka service discovery, achieving **80%** test coverage via JUnit and Cucumber.
- Built a full-stack resource management platform using React.js and PostgreSQL for personnel allocation and project tracking.

PROJECTS

Spectra-RAG | Go, Next.js, C++/Eigen, networkX, OpenRouter

Jun 2026

- Built a query-routing and retrieval system that makes small, resource-constrained LLMs viable over large knowledge graphs by recovering API controls (logit.bias, logprobs, frequency_penalty) stripped by free-tier providers.
- Implemented four application-layer algorithms in pure Go: a PCA→LDA query router (**85% LOOCV accuracy**), a Trie-based stream interceptor for canonical entity spelling, an SVD redundancy penalty, and a letter-vote evaluator. Measured on 22-question ablation: entity spelling **73.5%→81.4%**, groundedness **87.5%**.

WiFi DensePose: Privacy-Preserving Human Sensing | PyTorch, Deep Learning, Computer Vision

Nov 2025 – Present

- Developed a camera-free human pose estimation system that maps WiFi CSI data to dense UV coordinates, achieving **mAP > 0.65** in occluded environments.
- Built an end-to-end pipeline combining CNNs and Transformers with preprocessing for noisy wireless signals, then optimized inference for embedded GPUs to process live WiFi packets with under **50 ms** latency.

LLVM GPU Loop Tiling Optimization | C++, LLVM IR, CUDA

Mar 2025 – May 2025

- Implemented custom LLVM IR optimization passes for automated loop tiling in CUDA kernels, achieving a **3x speedup** on memory-bound GPU workloads by improving cache locality and memory bandwidth utilization.
- Built a Python benchmarking suite to measure optimization impact and analyze cache and memory-access behavior across kernels.

ML Parallelization Benchmarking Framework | Python, PyTorch, NumPy, scikit-learn

Jan 2025 – Mar 2025

- Built a benchmarking framework to measure parallel speedup, efficiency, and overhead for **4 ML** algorithms (PCA, KNN, Random Forest, Neural Network) across thread counts from **1 to 64** and multiple dataset sizes.
- Implemented algorithm-specific parallelism strategies: test-set chunking with ThreadPoolExecutor for KNN, parallel sub-forest training with estimator merging for Random Forest, and **nn.DataParallel** for GPU-distributed neural network training.

TECHNICAL SKILLS

Languages: Go, C++, Java, Python, Scala, JavaScript/TypeScript, SQL, CUDA, P4

Frameworks & APIs: Spring Boot, FastAPI, React.js, Next.js, Node.js, REST APIs, gRPC, Netflix Eureka

Cloud & DevOps: AWS (EC2, Lambda), Amazon EKS, Docker, Kubernetes, Terraform, Jenkins, CI/CD

Data & AI Infrastructure: Apache Spark, Kafka, Hadoop (HDFS), Hive, PostgreSQL, Redis, Pinecone, Neo4j, Vector DBs

ML & Retrieval: PyTorch, Transformers, LLMs, RAG, GraphRAG, LangChain, LangGraph, NLP, Distributed Training, Model Deployment

Tools: Git, Linux, Bash, Postman, Scapy, Selenium, Prometheus, Grafana, BMv2, p4c

EDUCATION

Virginia Tech

Blacksburg, VA

Master of Science in Computer Engineering; GPA: 3.64

Aug 2024 – May 2026

National Institute of Technology Karnataka

Surathkal, India

Bachelor of Technology in Electronics and Communication Engineering

Jul 2017 – May 2021